

Alec P. Wick

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EDUCATION

University of Nebraska Omaha	Omaha, NE
Master of Science in Data Science	May 2024
<i>Concentration:</i> Information Technology	GPA: 3.91
Doane University	Crete, NE
Bachelor of Science in Mathematics and Data Analytics	May 2021
<i>Minor:</i> Software Development	GPA: 3.62

EXPERIENCE

HDR Inc.	Omaha, NE
Data Scientist	May 2024 – Present
● Lead several strategic enterprise objectives, including an LLM-based marketing tool ○ Engineered dynamic prompt frameworks utilizing data from enterprise cloud platform ○ Presented to the executive leadership team (CEO, CFO, CTO, COO, CSO) ○ Mentioned at the company's annual meeting and highlighted in the strategic plan ○ 5,000+ uses within 6 months with a user base of 400 marketing staff ● Deployed and introduced financial predictive models for the Financial Leadership Team ○ XGBoost, LightGBM, FNN, Prophet, SARIMAX, Exponential Smoothing	
Business Intelligence Analyst	July 2021 – May 2024
● Partnered with finance department directors to provide effective reporting applications ● Deployed a regression-based model for procurement requisition and purchase order completion ● Experience managing and designing data architecture patterns for scalable enterprise use	
CATCH Intelligence	Omaha, NE
Data Science Intern & Apprentice	May 2019 – June 2021
● Trained a robust tree-based classification model with the capability to accurately predict fraudulent unemployment claims for clients during the COVID pandemic ● Led an internship cohort of 5 interns and mentored them throughout their summer internship	

RESEARCH

- Evaluating Trust in AI
 - Research examines the factors influencing human trust in AI within domains such as healthcare, law enforcement, public policy and employment
 - Utilizes machine learning to identify and rank variables such as age, income, and political ideology that influence trust in AI
 - Trained models predicted an individual's trust level with accuracy ranging from 73.3% to 86.7%, depending on the application domain

- Oregon Healthcare Evaluation
 - Fitted logistic regression models to predict emergency department utilization for a population of 30,000+ participants
 - Identified model covariates, including chronic illnesses (diabetes, asthma) and self-assessed pain levels as independent variables to predict healthcare utilization
 - Model predicted whether a patient would visit the emergency department with 64.4% accuracy based on their pre-existing health conditions
- Predicting Flight Delay
 - Extracted and cleaned 600,000+ flights using the Anyflight and dplyr package
 - Developed an applied predictive framework using Logistic Regression for binary classification and Multiple Regression for continuous variable forecasting
 - The classification model achieved an 83.33% accuracy rate while classifying flight delays within the validation set

SKILLS

- Deep technical expertise in Large Language Models (LLMs), including transformer architecture, training methodologies, deployment, inference and evaluation monitoring
- Leverage Python packages such as Pytorch, Tensorflow, Keras, and Scikit-learn to train, fine-tune and evaluate models
- Proficiently conduct data analysis and statistical modeling with expertise in Python and R
- Experience utilizing LaTeX and Quarto formatting to present professional research results
- Develop high-quality data visualizations to effectively tell a story through data using packages such as Matplotlib, Seaborn, Plotly, Plotnine, ggplot2 and Leaflet
- Design interactive reports and dashboards using PowerBI, Streamlit, Django, RShiny, Oracle BI, Tableau, ArcGIS Pro and SAP Predictive Factory

HONORS & AWARDS

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| ● 2 x HDR Corporate Pathfinder Award Winner | 2024 & 2025 |
| ● Jim Johnson Doane Mathematics Award | 2019 |

CERTIFICATIONS

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| ● Snowflake SnowPro Core Certification | 2025 |
| ● Doane Hansen Leadership Certification | 2021 |

PRESENTATIONS

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| ● HDR Technology Conference - Leveraging AI within Marketing | 2025 |
| ● HDR Finance Leadership Meeting - Effective use of Data & AI | |
| ● UNO Finance & Accounting Dinner - Extracting Insight from Data | |
| ● HDR Water Business Group Technology Update - AI in Marketing | |
| ● HDR Technology Special Topics - Utilizing AI within the Enterprise | 2024 |
| ● UNO Master's Project - Predicting Flight Delay | |